

Homework

SCENARIO 1: Emergency Communication

- (1) Problem: A major BTS has failed, causing loss of coverage, call drops and congestion on the remaining cells.
- (2) Role: Restore connectivity, protect emergency communication, and co-ordinate with power and reuse teams.
- (3) Assumptions:
 - Hexagonal cells, reuse factor $N=7$.
 - Neighbouring BTSs and the core network are working
 - COWs (cells on wheels) and satellites equipment are available.
- (4) Diagram:

(B2)(B3)

(B1)(X FAILED)(B4)

(B6)(B5)

+ cow inside the dead zone
- (5) Cell placement: Neighbours act as umbrella cells. priority goes to hospitals, shelters and police and fire stations.
- (6) BTS placement: Neighbours act as get up tilted antennas and more power.
- (7) Frequency reuse: Fractional reuse at cell edges
- (8) Handoff: Active calls drop and must be redialled.
- (9) Coverage/capacity: Coverage is partly restored, but weaker at the edges and indoors. capacity is the bigger problem, because neighbours absorb extra users and traffic spikes.
- (10) Challenges:
 - Call drops
 - Congestion
 - Interference
 - Power cuts
 - Backhaul damage
 - Emergency calls competing with public traffic
- (11) Solutions:
 - Traffic surge
 - Redundancy
 - Alternatives for emergency services
 - Priority
 - Restoration plan

conclusion:

Overlapping coverage, handoff, redundancy, and traffic prioritisation. Keep critical communication alive during a failure.

SCENARIO 2: Smart City

- (1) Problem: A city needs reliable coverage and capacity for millions of users and many IoT devices
- (2) Role: Design an efficient, scalable cellular architecture
- (3) Assumptions: $\rightarrow$ 4G/5G network
 $\rightarrow$ Fibre backhaul available
- (4) Diagram:
Macro cells (Hexagons, whole city)
+---- Micro cells (streets, roads)
+---- Small/pico cells (malls, stations, stadiums)
All BTSs $\rightarrow$ Fibre backhaul $\rightarrow$ core network $\rightarrow$ cloud/IoT platform
- (5) Cell selection: Downtown
Residential
Suburbs
Highways
- (6) High-capacity locations: Business district, railway/metro stations, bus terminals and traffic junctions.
- (7) BTS placement: Towers on rooftops in the city centre, street poles and lamp posts for micro/small cells and hospitals.
- (8) Cell sizes:

Type	Radius	Use
Macro	1 to 10km	Suburbs, wide coverage
Micro	200m to 1km	Streets
Pico	100 to 200m	Indoor hotspots
Femto	10 to 50m	Homes, offices
- (9) Frequency reuse: Reuse a cluster of $N=3, 4$ or 7 across the city
- (10) Interference sources: Co-channel interference b/w cells
- (11) Handoff: Vehicles
Pedestrians
Macro-micro layering

(12) Reduce congestion: Add more small cells
Traffic Shaping at peak hours.

(13) Small cells in dense areas: place them on poles and inside buildings to cover hotspots.

(14) Supporting IoT and mobile users: Mobile users
IoT

Conclusion:

A layered macro, micro and small-cell design with frequency reuse supports both people and devices.

SCENARIO 3: UNIVERSITY CAMPUS

(1) Problem: A campus of 2500 people has moderate usage normally but heavy peaks during exams, festivals and seminars.

(2) Role: Provide reliable coverage inside and outside for peak capacity.

(3) Assumptions: Mixed indoor and outdoor areas, fibre backbone, and wifi alongside cellular.

(4) Diagram:
[Hotels] --- (BTS 1) --- [Auditorium] --- (BTS 2) --- [Sports ground]

Cow for Events

(5) Continuous coverage areas: Auditorium, library, cafeteria, hotels in the evening.

(6) Highest capacity areas: classrooms, labs, library, hotels.

(7) BTS location: Rooftops of central buildings, near the hostel cluster, near the auditoriums.

(8) Cell arrangement: hexagonal pattern of medium cells outdoors.

(9) Frequency reuse: Reuse a cluster of 3 or 4 across the campus.

(10) Moving between buildings: The phone measures signal strength.

(11) Handoff: It keeps calls and data sessions continuous as users move. It uses signal thresholds and hysteresis.

(12) causes of poor indoor signals: concrete, steel, and metal-coated glass attenuation.

(13) Indoor solutions: DAS (Distributed Antenna system)

(14) Event traffic: Deploy cows near the event
temporarily add small cells

Conclusion:

Careful cell planning, indoor solutions and temporary capacity for events give reliable campus connectivity.

SCENARIO 4: RAILWAY NETWORK (200 KM)

- (1) Problem: Passengers need continuous connectivity along a 200 km route at high speed, through varied terrain.
- (2) Role: Provide uninterrupted coverage and manage fast handoffs.
- (3) Assumptions: The route is single line with sites along it, trains run at up to 120 to 160 km/h and 4G/5G is used.
- (4) Diagram:
city A == (BTS) == (BTS) == (BTS) == [Tunnel + repeater] == (BTS) == (BTS) == city B
← cell → ← cell → ← cell → ← cell →
- (5) Tower locations: Along the track at 2 to 5 km spacing at stations, near bridges and tunnel mouths and on hills in remote areas.
- (6) Cell arrangement: Linear chain of cells with directional antennas pointing along the track and overlapping edges for handoff.
- (7) Moving between cells: The phone keeps measuring the signal. As the train leaves one cell.
- (8) Handoff at high speed: The time in the overlap area is short, so handoff must be fast.
- (9) Causes of call drops: Late or failed handoff
Coverage gaps
Tunnels and cuttings
Doppler shift
- (10) Tunnels and obstacles: They block or attenuate signals. Metal train bodies add 15 to 25 dB of loss.
- (11) Remote area coverage: Tall towers on hills
Low-band frequencies (700/800 MHz)
- (12) Frequency reuse: A 1-D reuse pattern. Cells with the same frequency and separated by at least 2 cells.
- (13) Interference: Co-channels from reused frequencies
- (14) High-speed reliability: Redundant backhaul.

Conclusion:-

A linear cell layout, predictive handoff, and on-train relays keep services continuous at speed.

SCENARIO 5: SHOPPING MALL

(1) Problem: Weak signal and slow internet indoors, with 30,000 visitors a day and peaks on weekends & festivals.

(2) Role: Improve indoor coverage and capacity.

(3) Assumptions: A multi-floor building, with fibre available and several operators sharing the infrastructure.

(4) Diagram:

Roof top macro (outdoor)

Floor 3: [SC] --- [SC] --- [SC]

Floor 2: [SC] --- [SC] --- [SC]

Floor 1: [SC] --- [SC] --- [SC]

All → Baseband unit (BBU) in the server room → Fibre → Core network

(5) Reasons for weak signals: Distance from towers
Large crowds

(6) Building materials: concrete and brick absorb signals, metal reflects or shield them.

(7) Better coverage needed: Atrium, corridors, escalators and lifts, parking and storage areas.

(8) High capacity needed: Food court, cinema, atrium and Events area, anchor stores, counters.

(9) Indoor equipment locations: Ceilings of each floor, near atriums and the food court.

(10) Small cells: They bring the signal closer to users, improve signal quality.

(11) Interference: Wi-fi and Bluetooth in shared bands
Signal leakage through windows

(12) Weekends and festivals: Wi-fi offload and Wi-fi calling
Load balancing

(13) Moving between floors: Each floor has its own cells. cells overlap near stairs, escalators and lifts.

(14) Coverage vs Capacity:

	<u>coverage</u>	<u>capacity</u>
Goal	Signal Everywhere	Support many users
Drivers	Area and obstacles	No. of users and data demand
Solution	Antennas, DAS, Repeaters	More cells, spectrum, MIMO
Where	Basements, lifts, Corridors	Food court, atrium, cinema

Conclusion:

A mall needs both: good coverage Everywhere through DAS or small cells, plus dense capacity in crowded zones.